# Satisfeed Final Report

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## Description

Satisfeed is a nonprofit that provides hunger relief for families in and around Gwinnett County through food co-ops and weekend programs. Their mission is to fight food insecurity, reduce food waste, and connect communities. To more efficiently achieve their goal, Satisfeed uses a data-driven dashboard to visualize demographics and service usage, providing the organization with insights for program planning and resource allocation. In the spring of 2026, our team worked with Satisfeed to enhance their existing dashboard by providing additional data and studies to boost operational efficiency and provide a much clearer understanding of how to combat food insecurity.

## The Team

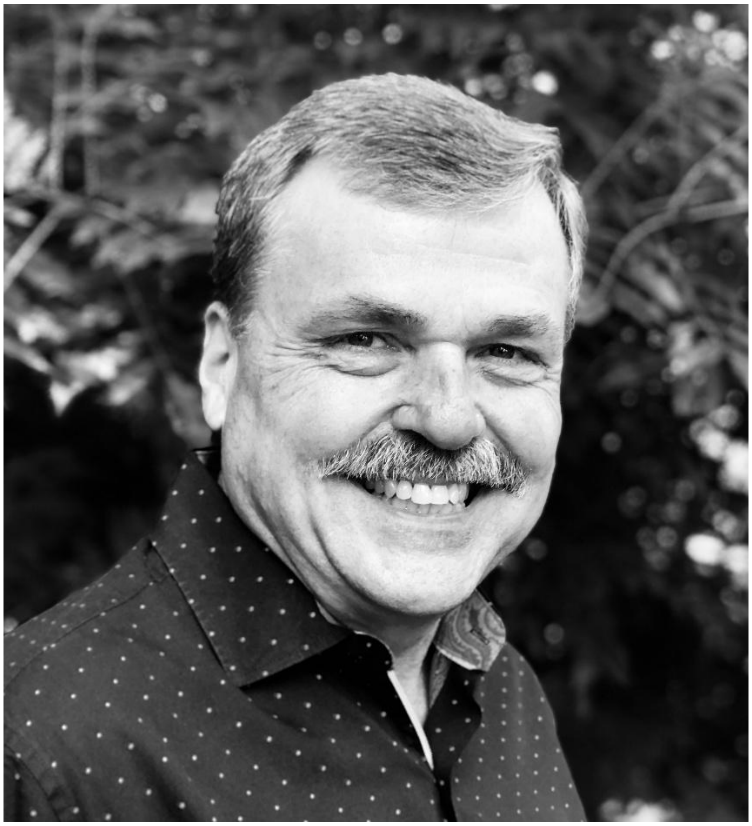
**Jaylan Igbinoba** - Visualization/Project Documenter

**Ethan Boderas** - Data Modeler/Client Liaison

**Wilmer Cifuentes** - Data Analyzer/Project Manager

## The Client

The client for this project is Tim Turner, founder & CEO of Satisfeed.



## Team Plan

The project unfolded in three stages: first, we gathered rich data on food insecurity and its wide-ranging effects. From there, we dove into analysis — building visualizations and running algorithmic tests to find meaningful insights. Finally, we brought it all together by transforming the cleaned datasets into a geographic dashboard that provides vital information at a glance.

## Technologies Used

- Jupyter Notebook
- Pandas
- Matplotlib
- Principal Component Analysis (PCA)
- Standard Scaler
- K-means Clustering
- Correlation Analysis
- Streamlit

## Flow Chart Of The Project

TODO

## Collections

Link to the website with the School population and the Free and Reduced Lunch data: https://www.decal.ga.gov/SFSP/Eligibility.aspx?utm

Link to website containing the SAT, and School Grads Dataset: https://gosa.georgia.gov/dashboards-data-report-card/downloadable-data

Link to Feeding America dataset: https://map.feedingamerica.org/county/2023/overall/georgia

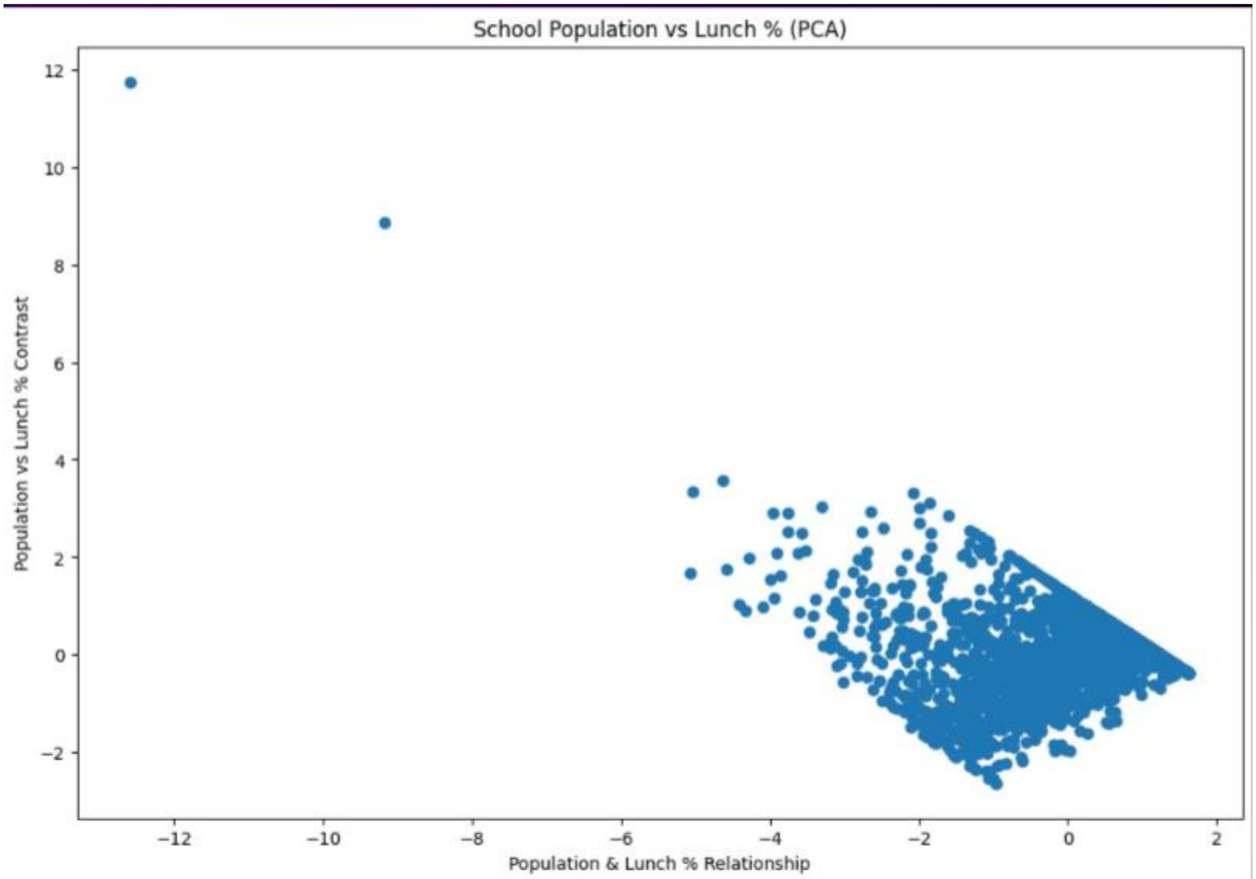
## Methods Used to Clean the Datasets

Across the various datasets, cleaning primarily involved dropping irrelevant or redundant columns, filtering rows to focus on specific regions or timeframes, and renaming columns for improved readability. Additionally, some datasets had null values dropped, while others involved extracting and reorganizing specific information into new columns.

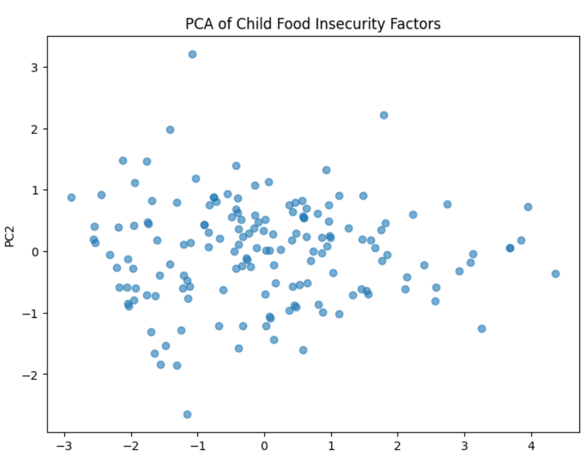
## Methods Used to Analyze Data & Results

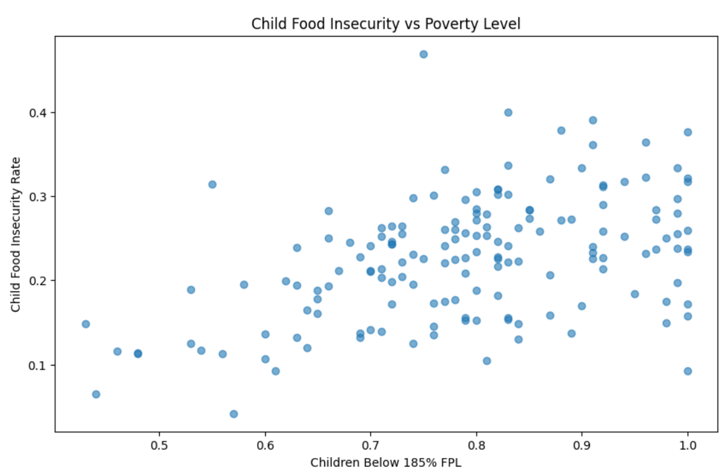
### Jaylan

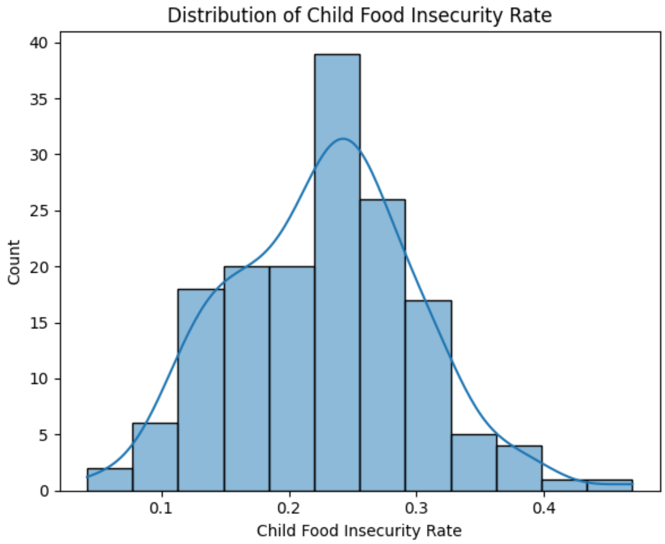
For my dataset, I used the PCA method to try and figure out if my hypothesis is true or not. My hypothesis was that schools with a higher population will have a higher Free & Reduced Lunch percentage. My results found out that the two variables have no correlation with each other at all. You can tell this is the case because most of the Data points fell under the -2 to 2 marks which will indicate no correlation.

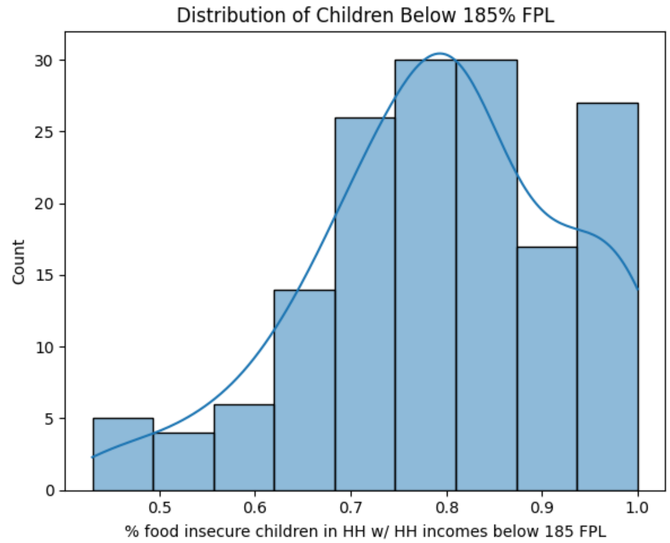

School Population vs Lunch % (PCA)
Population vs Lunch % Contrast
Population & Lunch % Relationship
12
10
8
6
4
2
0
−2
−12
−10
−8
−6
−4
−2
0
2

# Wilmer


PCA of Child Food Insecurity Factors
PC2
PC1
3
2
1
0
−1
−2
−3
−2
−1
0
1
2
3
4


Child Food Insecurity vs Poverty Level
Child Food Insecurity Rate
Children Below 185% FPL
0.4
0.3
0.2
0.1
0.5
0.6
0.7
0.8
0.9
1.0

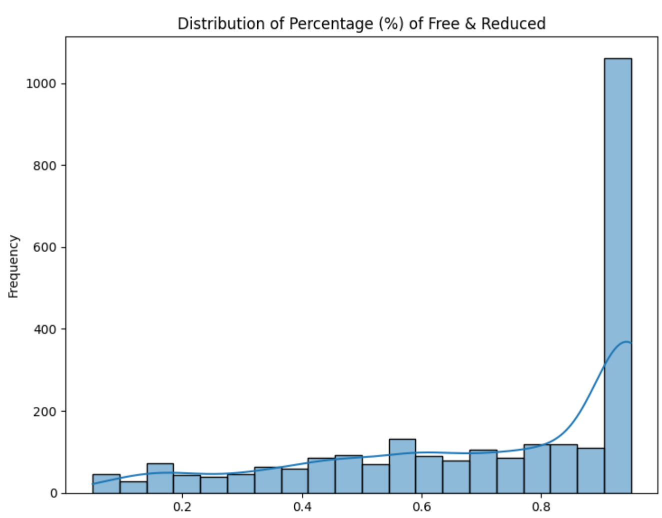


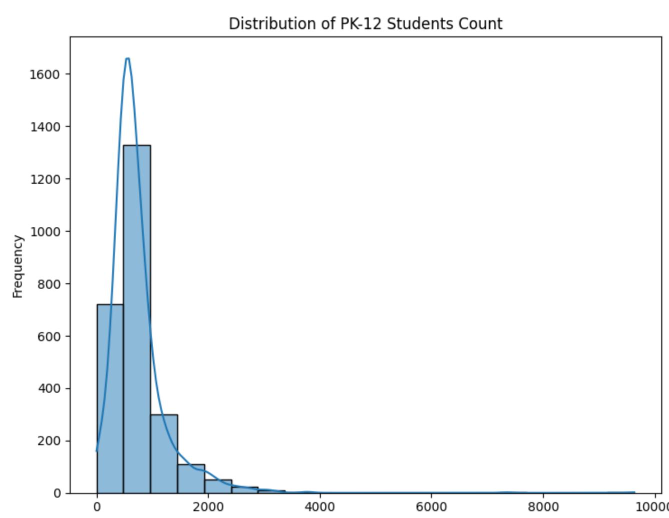



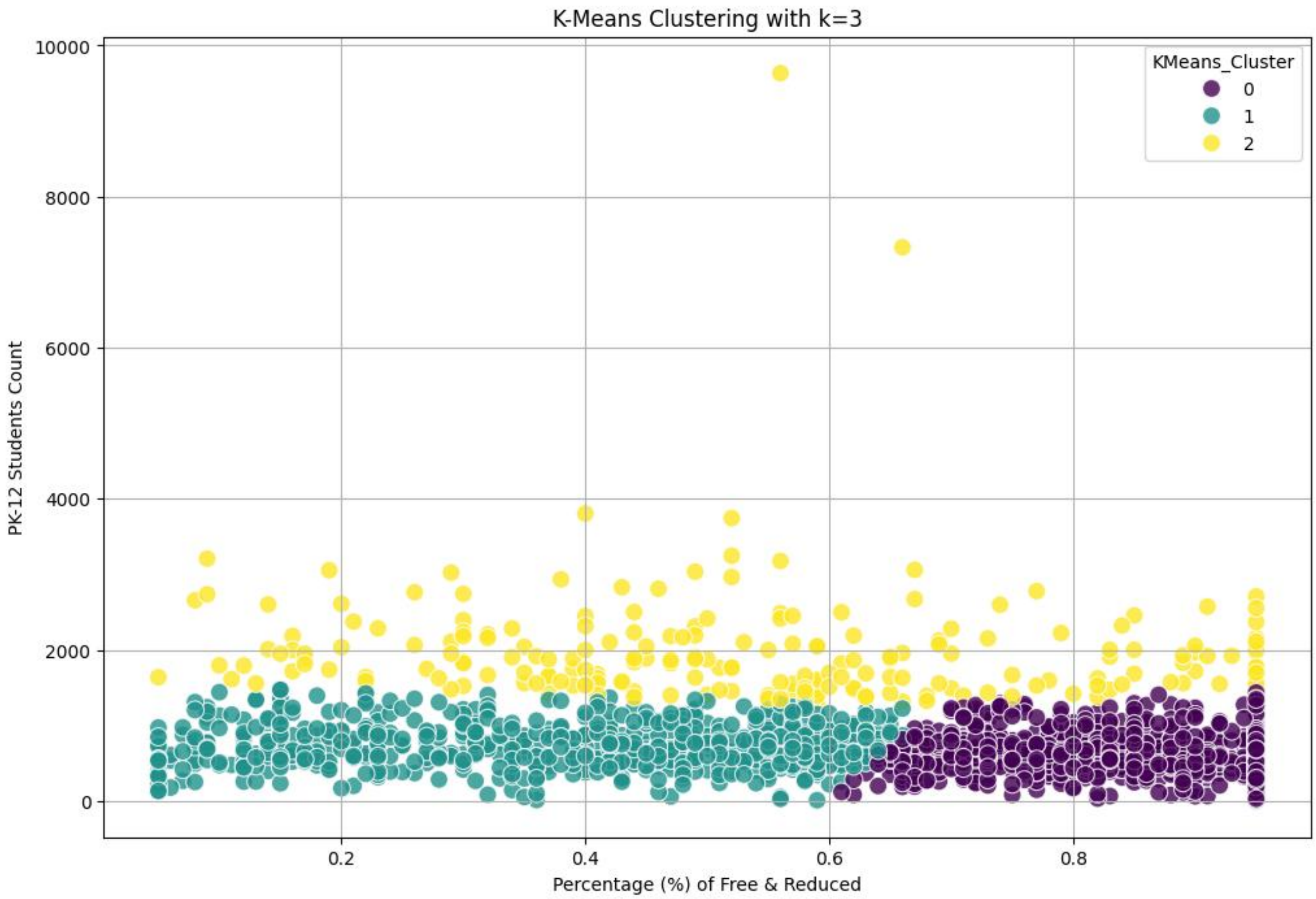

The PCA results show that the first principal component explains about 77% of the total variance, meaning most of the important information in the dataset is captured in a single dimension. This indicates a strong relationship between child food insecurity and poverty related factors, suggesting that these variables tend to move together across counties.

The distribution plots provide additional insight into these patterns. The child food insecurity rate appears to follow a roughly normal distribution, showing variation across counties but remaining relatively centered. In contrast, the percentage of children living in households below 185% of the Federal Poverty Level is skewed toward higher values. This suggests that many counties have a large proportion of low-income children, which helps explain the higher levels of food insecurity observed in those areas.

## Ethan

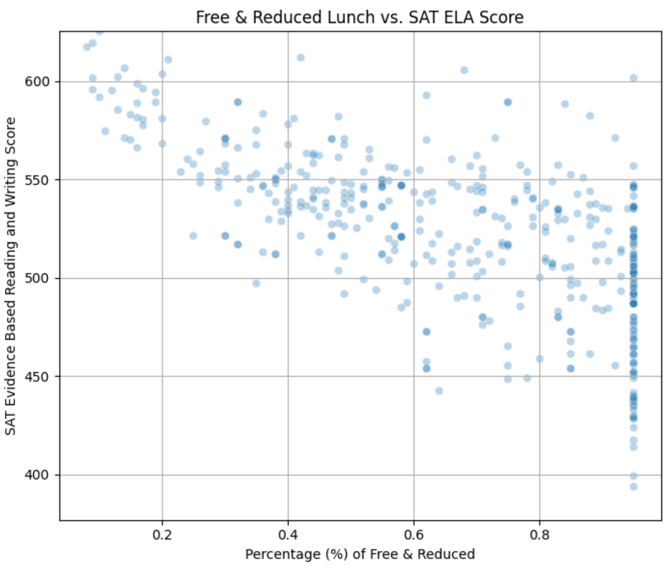


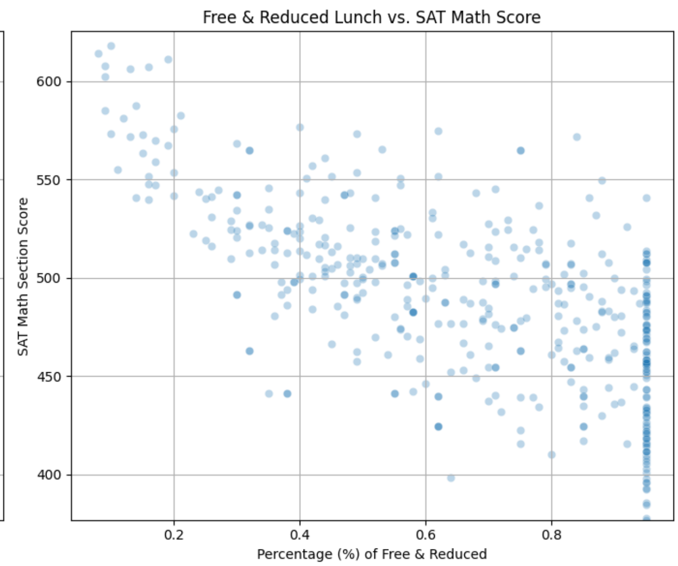



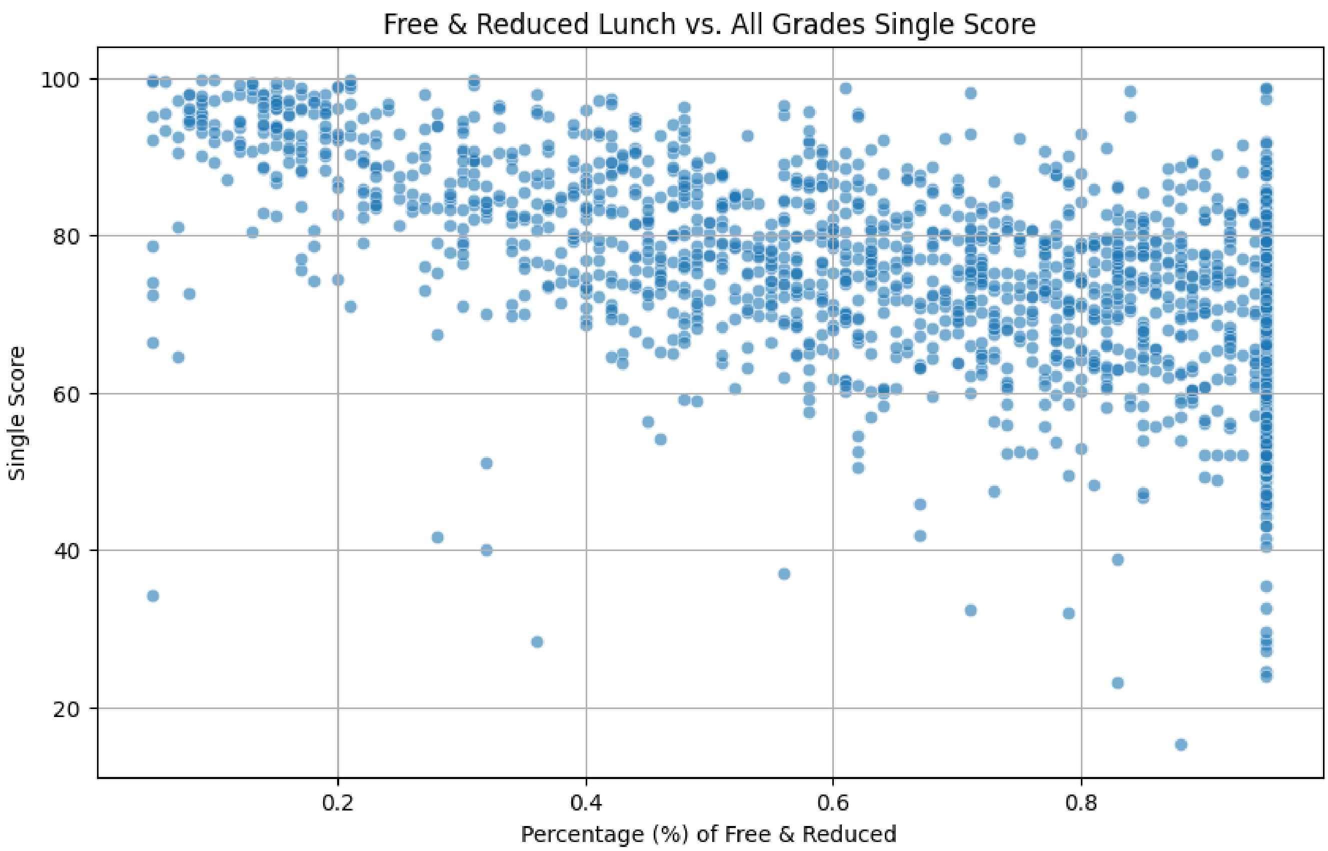


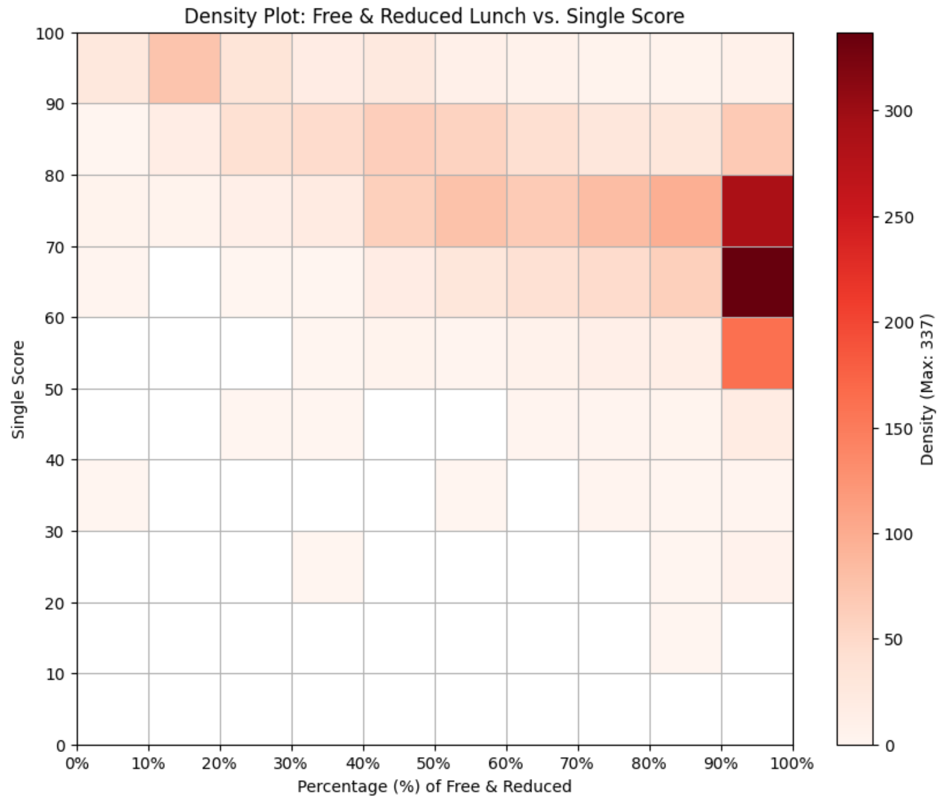

The above investigates the hypothesis that "Schools with a higher population will have a higher Free & Reduced Lunch percentage." Using K-means clustering on 'Percentage (%) of Free & Reduced' and 'PK-12 Students Count', I identified three optimal clusters via the Elbow Method to examine the relationship between student population and lunch eligibility.

The results did not support the hypothesis. Cluster 0 showed high percentages of free and reduced-price lunch but wide variance in student count, while Cluster 2 — comprising larger schools — displayed a broad range of lunch eligibility percentages. Together, these findings suggest that student count does not directly correlate with free and reduced-price lunch rates and may, in fact, indicate an inverse relationship, with smaller schools sometimes showing higher eligibility proportions.


Free & Reduced Lunch vs. SAT ELA Score
SAT Evidence Based Reading and Writing Score
Percentage (%) of Free & Reduced


Free & Reduced Lunch vs. SAT Math Score
SAT Math Section Score
Percentage (%) of Free & Reduced


Free & Reduced Lunch vs. All Grades Single Score
Single Score
Percentage (%) of Free & Reduced




Hypothesis 3 — "Schools with a higher percentage of free and reduced lunch will have lower SAT scores or Single Scores" — was tested using two academic performance metrics. The Single Score is an overall performance rating (0–100) for public schools, calculated by the Governor's Office of Student Achievement (GOSA). To conduct the analysis, three datasets were merged to link SAT scores, Single Scores, and free- and reduced-lunch eligibility. Correlation coefficients, scatter plots, and a density plot served as the primary analytical methods.

The results strongly supported the hypothesis. SAT scores showed a moderate-to-strong negative correlation with lunch eligibility, with the coefficients of approximately -0.65 for Evidence-Based Reading and Writing and -0.67 for Math. The Single Score analysis yielded a similar coefficient of -0.60. Scatter and density plots visually confirmed this trend across both metrics, suggesting that socioeconomic factors — as reflected by free and reduced lunch eligibility — are significantly associated with academic outcomes. This strongly suggests that hunger negatively impacts student performance.

## Iteration Summaries

### Iteration 1 – This iteration focused on finding relevant and helpful data on food insecurity. The data collected consists of student grades, food insecurity levels, school ethnicity percentages, and school free & reduced-price lunch percentages.

Iteration 2 – With the data collected and cleaned, we moved into analysis. We developed visualizations grounded in our hypotheses and then tested them through clustering and correlation analyses.

Iteration 3 – In the final segment of the project, we examined the data, identified key insights with direct implications for Satisfeed's operations, and then displayed them on an interactive geographic dashboard.

## GitHub Repository Link

This link contains all the work that the current and former teams worked on: Link

The GitHub repository contains all datasets, along with full documentation of their analysis. It also includes the final cleaned datasets used for the dashboard, as well as all website and dashboard assets.

## Website & Dashboard Link

Link to website dashboard

## Notebook Link

Link to Notebook

## Jira Dashboard

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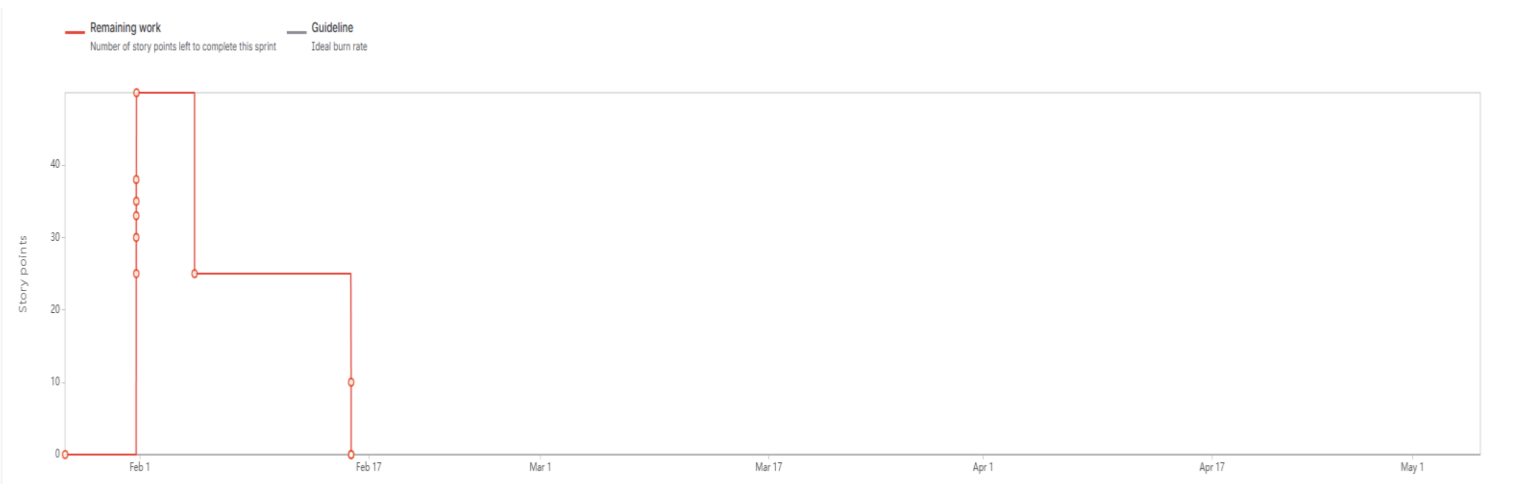
# Burndown and Velocity charts

## First Iteration - Velocity Chart


## First Iteration - Burndown Chart




## Second Iteration - Velocity Chart

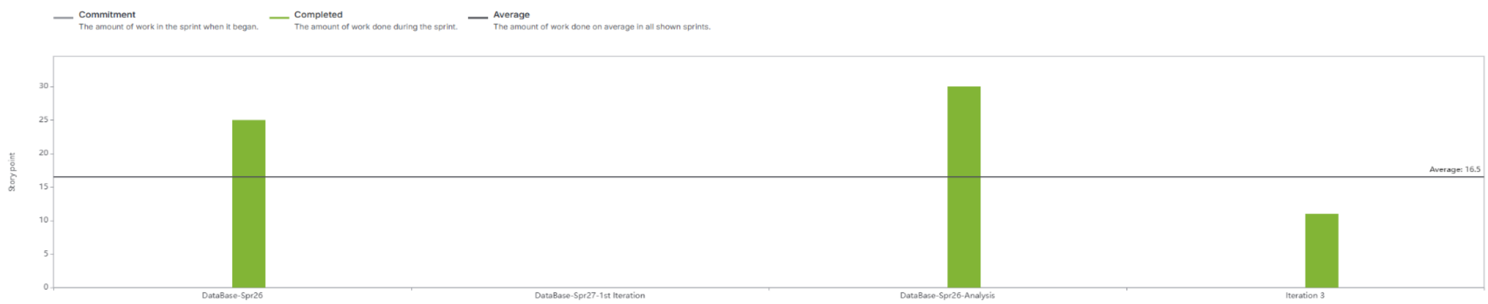
# Second Iteration - Burndown Chart


# Third Iteration - Velocity Chart

Spaces / Food Delivery / Reports

**Velocity report**

How to read this report




| Sprint | Commitment | Completed |
|---|---|---|
| DataBase-Spr26 | 0 | 25 |
| DataBase-Spr27-1st Iteration | 0 | 0 |
| DataBase-Spr26-Analysis | 0 | 30 |
| Iteration 3 | 0 | 11 |

# Third Iteration - Burndown Chart

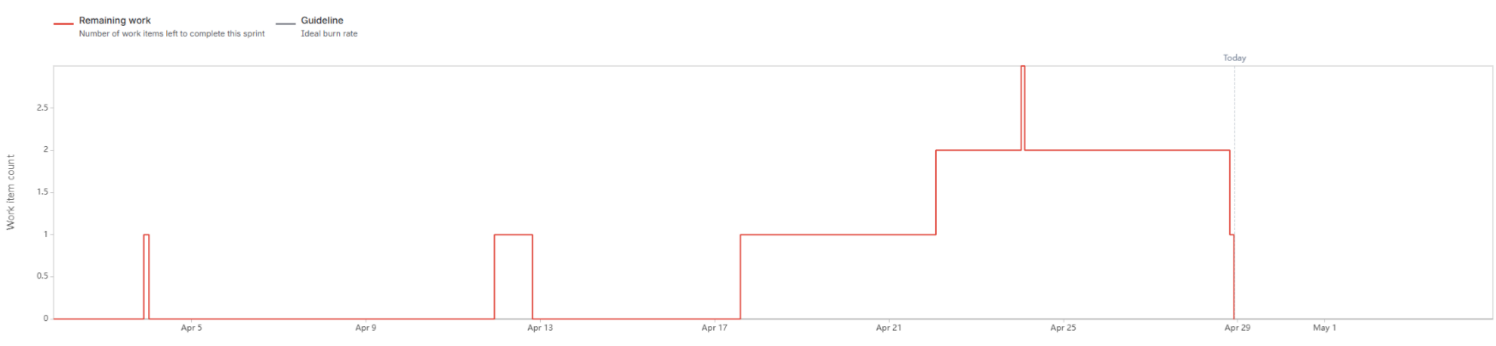
**Sprint burndown chart**

How to read this report

Sprint: Iteration 3

Estimation field: Work item count

**Date** - April 1st, 2026 - May 4th, 2026



# Project Demo

text

# Features Implemented

### Website:

- Project introduction
- Directory side panel
- Direct GitHub Repo button
- Direct GitHub fork button
- Ability to change background to light or dark mode
- Webpage loading indication at top right

### Dashboard:

- Geographic map
- Able to move and zoom the map
- School clustering with count
- Food insecurity color indication
- School data is shown when selecting a school on the map

# Known Issues

- Dashboard map is somewhat slow
- Dashboard demographic category percentages are not being read correctly

# TODO Tasks

- The client requested the ability to view which libraries have a kiosk that allows people to sign up for food services
- The dashboard can be greatly enhanced with a filtering function
- The dashboard data should be updated to maintain relevance